INCH-POUND

A-A-52480
October 29, 1993
SUPERSEDING
(see note d)

COMMERCIAL ITEM DESCRIPTION

CONTROL ASSEMBLY, PUSH-PULL

The General Services Administration has authorized the. use of this commercial item description (CID) as a replacement of MIL-C-62191, MS500043B, MS500044C, MS500045C, MS500073B, and MS500074B which are inactive for new design.

ABSTRACT

This CID covers push-pull control assemblies, of the single core wire type, for general applications, used to regulate and actuate many internal combustion engine and vehicular accessories, such as: throttles, idle controls, chokes, defrosters, and governors.

a. Classification. Push-pull control assemblies shall be of the following types (see salient characteristic (SC) c):

	Mechanisms
Type I	- Friction lock... (figure 1)
Type II	- Ratchet lock . . . . (figure 2)
Type III	- Vernier adjust. . (figure 3)
Type IV	- Turn-to-lock . . . . (figure 4)

SALIENT CHARACTERISTICS.

a. Materials. The handles and cover plates shall be made of molded plastic or zinc alloy casting. Plunger rods, guide tubes, core wires, and conduits shall be made of stainless steel or steel that has been plated with a corrosion-resistant electro-chemical finish. Guide tubes may also be made of brass, zinc,

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AMSC N/A FSC 2990

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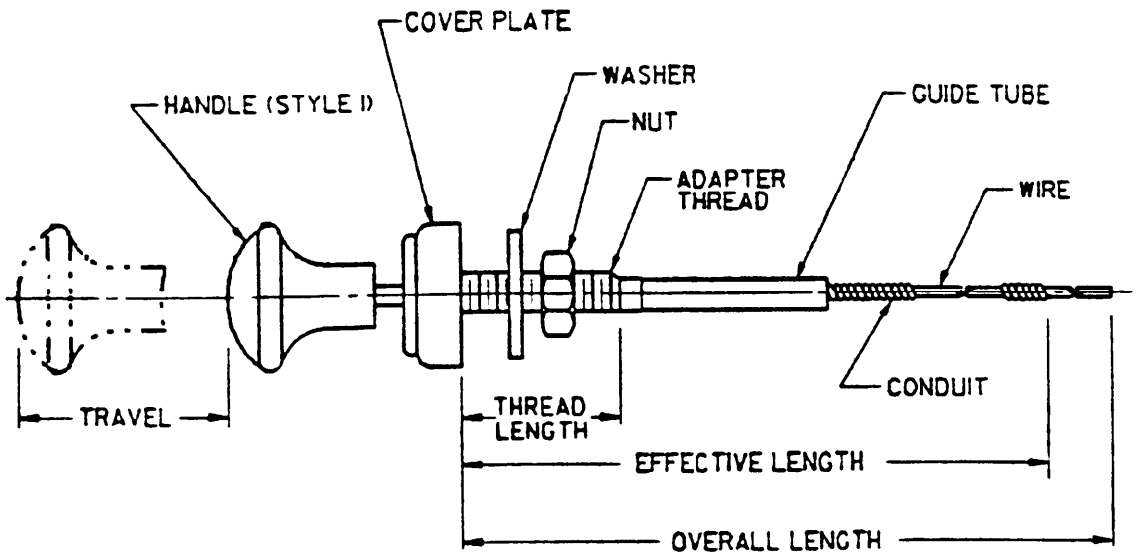


FIGURE 1. Type I - Friction lock.

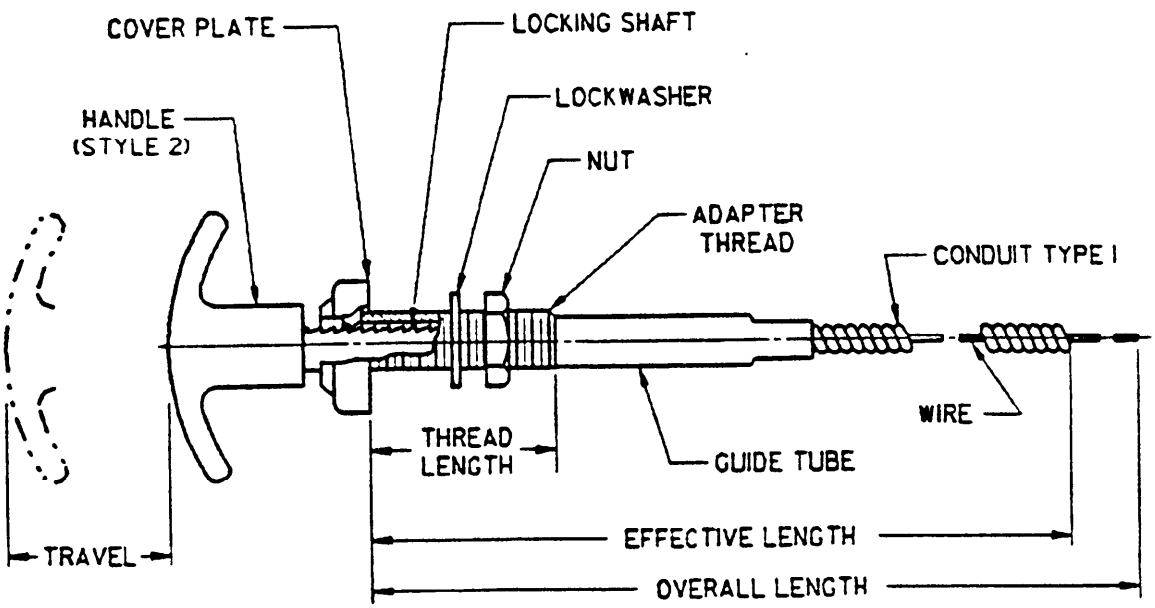


FIGURE 2. Type II - Ratchet lock.

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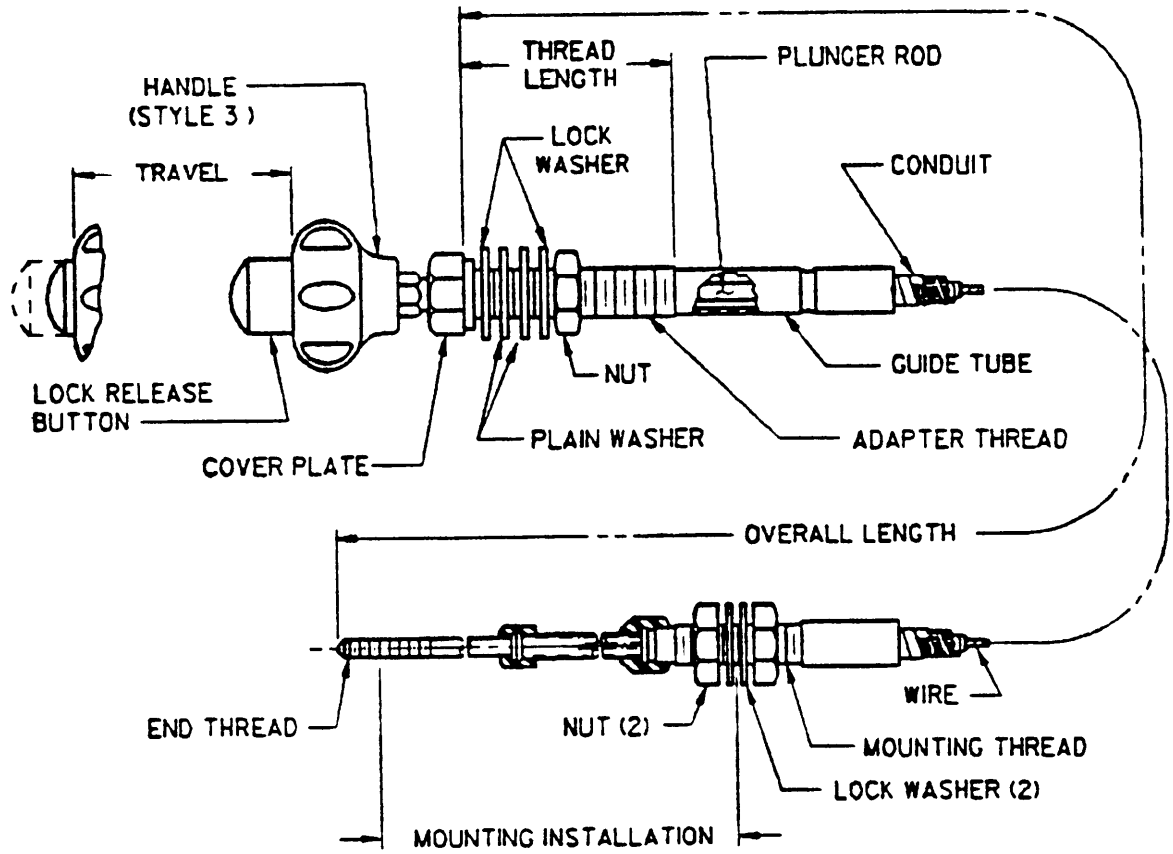


FIGURE 3. Type III - Vernier adjust.

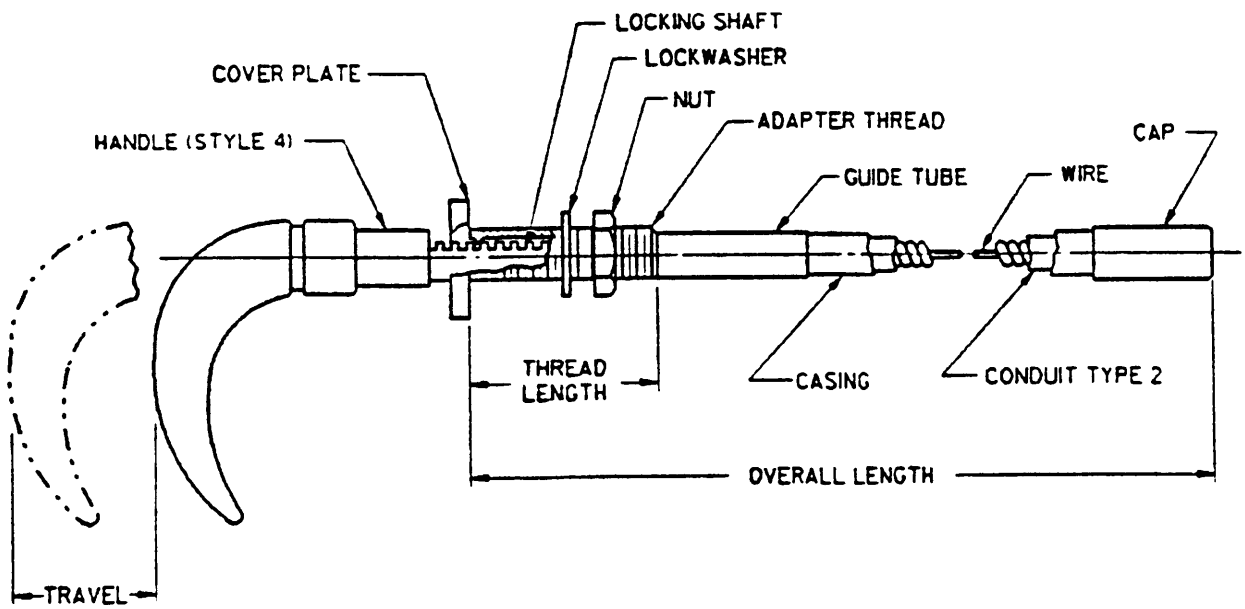


FIGURE 4. Type IV - Turn-to-lock.

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aluminum, or some combination of these metals with steel. The use of recovered materials made in compliance with regulatory requirements is acceptable providing that all requirements of this CID are met (see note e).

b. Design and construction. The general design and construction features of the push-pull control assemblies are shown in figures 1 through 4. The four assembly types consist primarily of various styles of control handles attached to various locking mechanisms and cable assemblies. A plunger rod shall permanently and rigidly attach the handle to a core wire which can be actuated backwards and forwards inside a flexible conduit. The conduit shall be composed of galvanized steel in the form of a closed helix. When specified (see note b), the conduit shall be provided with a plastic inner lining or encased in a plastic or rubber sleeve. The plunger rod shall be able to slide freely inside a threaded mounting adapter that is attached to a guide tube. Mounting hardware shall be provided with each assembly. Detailed and specific features and dimensions will be determined according to specific applications and specified in an engineering drawing (see note b).

1. Configuration. Many of the general design features (handles, cover plates, conduit types, etc.) shown in any of the four figures may be installed on a different assembly type. Handle style 2 (see figure 2) may have a horizontal or vertical locking position.

c. Performance. All assemblies (figures 1-4) shall be capable of operating satisfactorily in a temperature range of -65 to +115 degrees Fahrenheit, with up to 30 pounds of push-pull actuating force. Unless otherwise specified (see note b), control handles shall withstand the steady, gradually applied pull load that is given for each assembly type without separation from or damage to the assembly. They shall operate smoothly throughout the entire travel length without binding. The conduits shall show no permanent set from a bend radius of 3 inches.

1. Friction lock (Type I). The Type I assembly (figure 1) shall require 3 to 8 pounds to adjust and be able to withstand a maximum pull load (see SC c) of 120 pounds. With a conduit bend radius of 3 inches, the assembly shall function smoothly with up to a 20 pound push-pull actuating force.

2. Ratchet lock (Type II). The Type II assembly (figure 2) shall possess the capability of being pulled outwards into locked positions, at not more than 0.125 inch intervals, without turning the handle. Unlocking is necessary only to actuate inward and shall require a 1/4 to 3/4 turn of the handle to the right or left. The assembly shall withstand a maximum pull load (see SC c) of 75 pounds. With a conduit bend radius of 3 inches, the assembly shall function smoothly with up to a 20 pound push-pull actuating force.

3. Vernier adjust (Type III). The Type III assembly (figure 3) shall use a push-button lock release mechanism to provide coarse adjustment throughout the entire travel range and a knob for fine adjustment to the coarse setting. This vernier adjustment shall automatically become operable when the coarse adjustment button is released. The assembly shall withstand a maximum pull load (see SC c) of 75 pounds. With a conduit bend radius of 3 inches, vernier adjustment shall be capable with a maximum force of 10 inch-pounds.

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4. Turn-to-lock (Type IV). The Type IV assembly (figure 4) shall possess the capability of push-pull actuation throughout its entire travel length until it is locked into position with a 1/4 to 3/4 turn of the handle to the right or left. The locking positions shall be set at not more than 0.125 inch intervals. The assembly shall withstand a maximum pull load (see SC c) of 75 pounds. With a conduit bend radius of 3 inches, the assembly shall function smoothly with up to a 20 pound push-pull actuating force.

5. Lock holding ability (Type II, III, and IV). These assemblies shall not allow core movement of more than 0.031 inch after application of an 8 ounce load in alternating directions along the horizontal axis of the core, away from and towards the control handles, for a complete test cycle of 50 times in each direction. There shall also be no change of pre-set control adjustment positions nor damage to the control assemblies.

d. Surface treatments. Unless otherwise specified (see note b), handles and cover plates shall be painted green 383 according to the Army drawing 12369000, Chemical Agent Resistant Coatings (CARC) Paint Systems Index.

e. Identification and markings. The product shall be identified and marked in accordance with the engineering drawing to include, as a minimum, the military part number (see note c) and the manufacturer's CAGE code and part number (see note b). When imprints are specified, they shall be permanently indented into the handles with white enamel at a height of 0.125 inch minimum. When specified (see note b), guide tubes shall be pre-lubricated with grease that can withstand the operating temperature range specified (see SC c). Hot stamping is acceptable on plastic knobs.

QUALITY ASSURANCE PROVISIONS

a. Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

b. Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this CID and that the product conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

PRESERVATION, PACKAGING, PACKING, LABELING, AND MARKING

Preservation, packaging, packing, labeling, and marking for the desired level shall be as specified in the contract (see note b).

NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

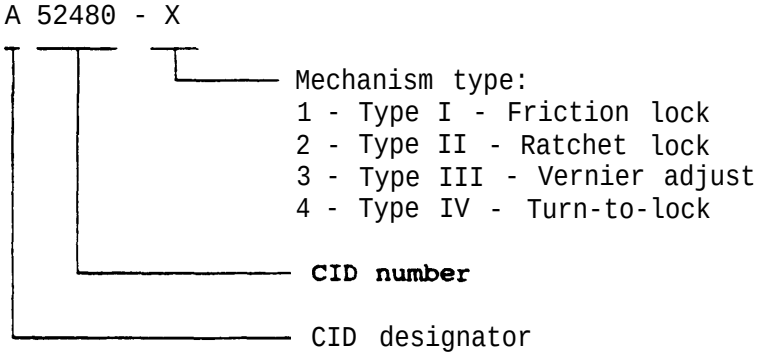
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a. Addresses for obtaining copies of Army drawing. Copies of Army drawing 12369000, Chemical Agent Resistant Coatings (CARC) Paint Systems Index, are available from the Contracting Officer, U.S. Army Tank-Automotive Command, Warren, MI 48397-5000.

b. Ordering data. Acquisition documents must specify the following:

1. Title, number, and date of this CID.
2. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced.
3. Number and date of the engineering drawing.
4. Whether an inner lining or an outer covering is required.
5. Maximum pull load, if different.
6. Color of paint, if different.
7. If a pre-lubricated control is required.
8. Part number and quantity required.
9. Selection of applicable level and packaging requirements.

c. Part or identification number (PIN). The PINs to be used for the push-pull control assemblies acquired to this CID is created as follows:



d. Supersession and cross-reference data. This CID is interchangeable/substitutable with, and supersedes MIL-P-62191 dated 25 March 1974; MS500043B, MS500073B, MS500074B dated 29 April 1975; MS500044C dated 25 September 1977; and MS500045C dated 9 January 1981. These documents are inactivated to support acquisition of replacement items for existing designs and equipment.

e. Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practicable.

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MILITARY INTERESTS:

Custodians
Army - AT
Navy - YD
Air Force - 99

Review Activities
Army - ME
Air Force - 11, 82
DLA - CS

User Activity
Navy - MC

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FSS

PREPARING ACTIVITY:
Army - AT

(Project 2990-0112)

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